Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “current” to explain your answer. Sentence starters are optional.

series: connected along one unbranched current path

current: rate of charge flow

resistance: voltage divided by current for a stated operating condition

01 Warm-up: bring back an earlier idea

Explain what a potential difference of **5 V** means in joules per coulomb.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

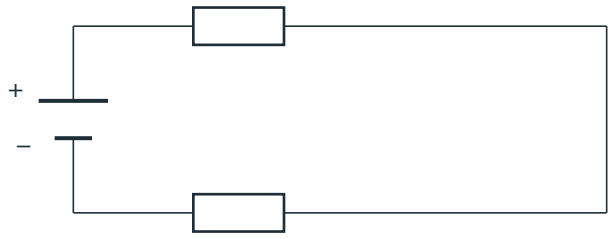
02 Worked example: explain the prediction

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$. Series resistors share one unbranched current path.

Worked reasoning: Two series resistors **2** and **4 ohms** on a **12 V** ideal supply have total **6 ohms** and current **2 A**. Drops are **4 V** and **8 V**; they add to **12 V**.

Your turn: Draw the loop and **label current** and both voltage drops.

Sentence starter: In my work, I used … to show … . I labeled it … .



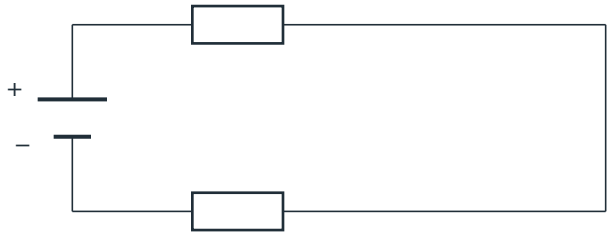
Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Trace the only current path in the series diagram. If either resistor connection opens, **what happens** to the current?

Sentence starter: In my work, I used … to show … . I labeled it … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

Series circuits

antocciphysics

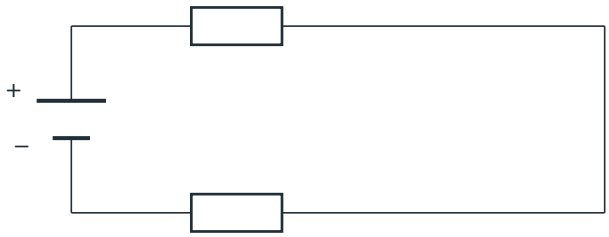
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **6 V** battery powers **2 Ω** and **4 Ω** resistors in series on a circuit board.
1. **Label both resistors** and the battery.
2. **Add resistances**, then **find the current through each** resistor.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student assigns more current to the smaller series resistor. **Explain why this would imply** charge accumulating between resistors.

Sentence starter: Both ... . They differ in ... because ... .

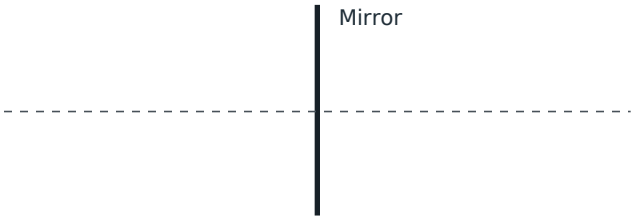
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A light ray strikes a plane mirror at **25°** from the normal. **Find the reflection angle** and **label both angles** on a sketch.
Draw the rays and **label both angles** from the normal.

Word help: reflection = a wave returning from a boundary

Sentence starter: My design includes ... . It does what is needed by ... .



Add object / image or incident / reflected rays.

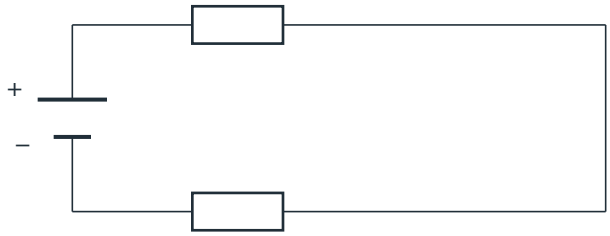
Marzano check - circle: 1 2 3

07 Exit check

In a **9 V** series loop, one component drops **3 V**.
1. **Find the other drop** if wires are ideal.
2. **Label the circuit** and **show the current path or** paths.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.



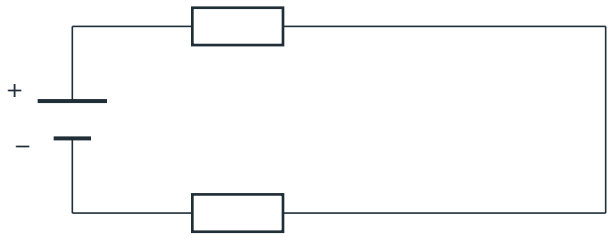
Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Choose two positive series resistances totaling **10 ohms**. On a **10 V** supply, find current and each voltage drop. **Give a second valid design**. **Label the circuit** and **show the current path or** paths.

Sentence starter: My design includes ... . It does what is needed by ... .



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3